# Phases of research process in scientific research

## Biostatistics and Scientific Research

2-3-2026

Dr.Lama Rafieh

Second semester-Third year

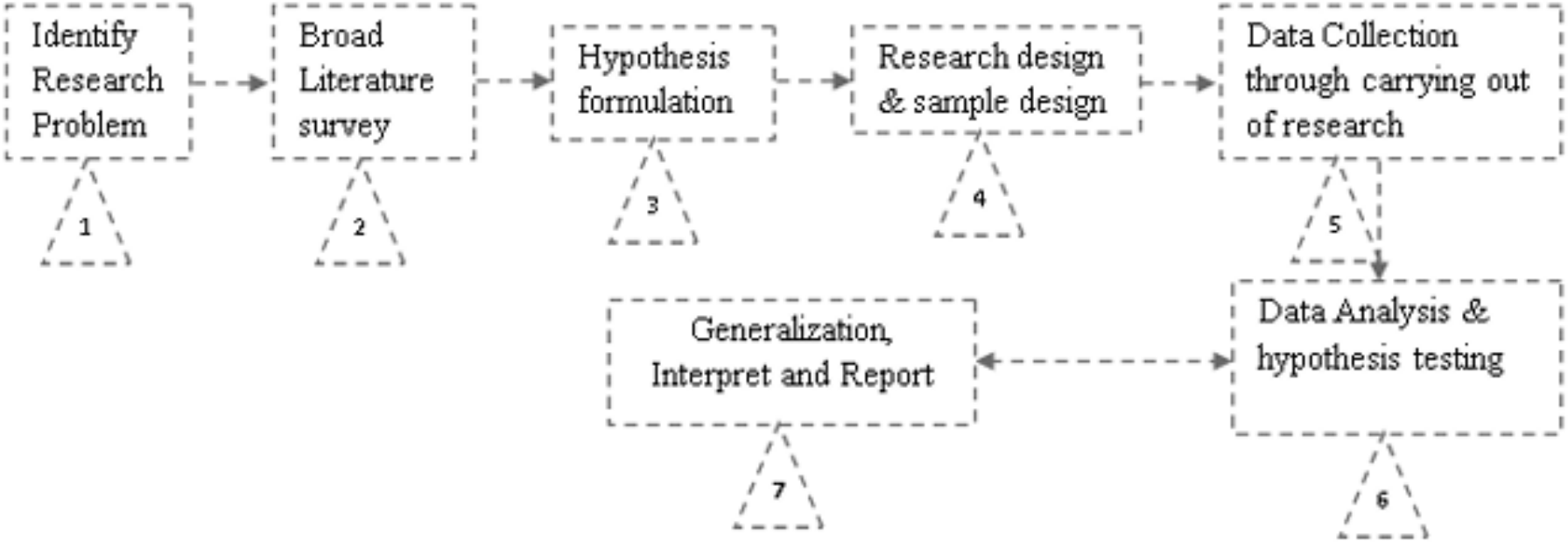
Week 2

# Outlines

- Phases of research
- Definition of journal (original) articles
- Title of the article
- Authors names and affiliation
- Keywords

# This chart represents a research process.

**Fig. 1: Flow chart of research process**



# Research process

- The figure shows that the research process having a number of closely related actions, as shown from step 1 to 7.
- But these activities should be following in a strictly prescribed sequence otherwise researcher may face the problem in completion of the research.
- In the research process, each step is specific and they are separate and distinct from each other.

# Steps regarding the research process

- Identification of research problem
- Broad literature survey
- Hypothesis formulation
- Preparation of research design
- Determining sample design
- Data collection
- Analysis of data
- Hypothesis testing
- Generalizations and interpretation
- Preparation of the report or presentation of the results,

# Journal (original) articles

- **Definition:**

An original article is a type of research paper that:

- Presents new, **original** research findings or results
- Presents new, **unpublished** research findings or results
- Presents reports on original research conducted by the **authors**
- Contributes to the existing body of knowledge in **a particular field** or discipline.

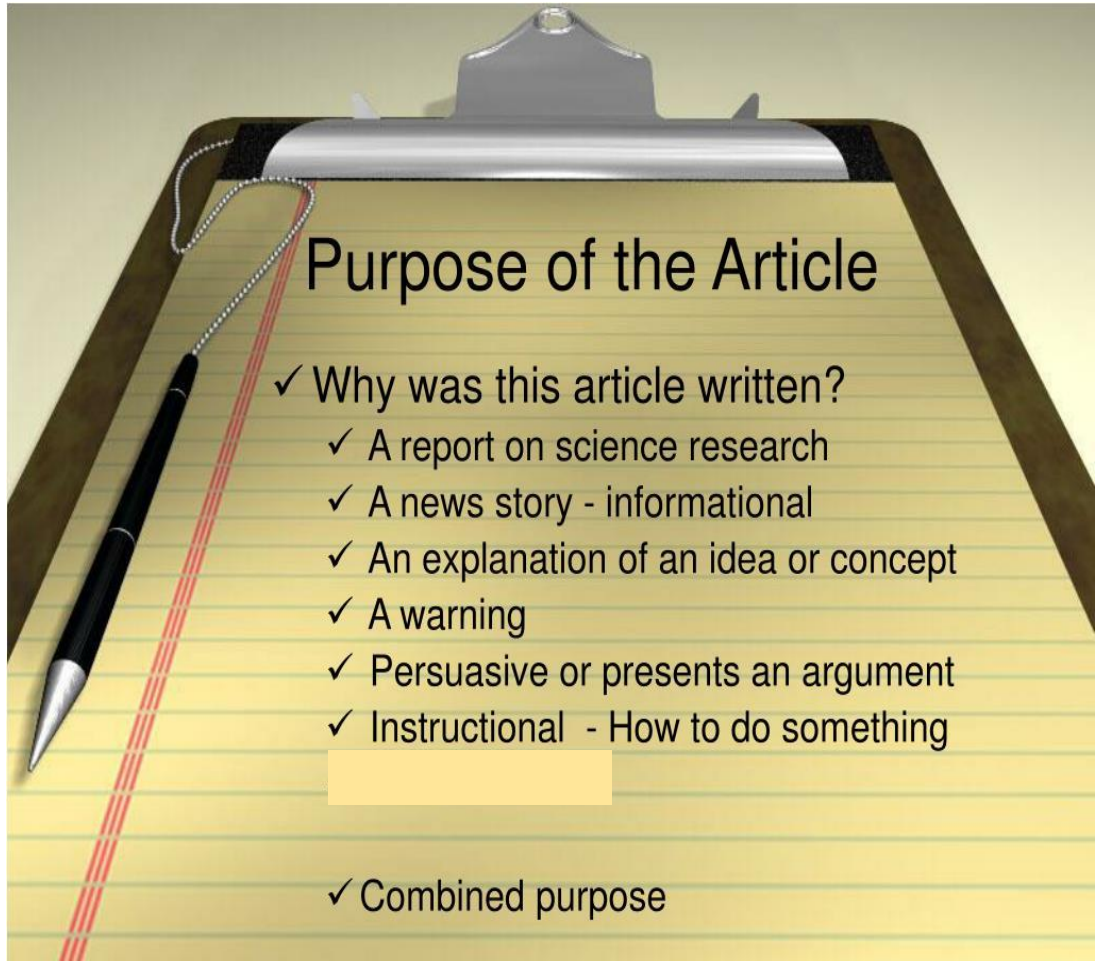
# Journal (original) articles

- **The primary purpose of an original article is to:**

1. **Contribute** to the advancement of knowledge in a particular field or discipline

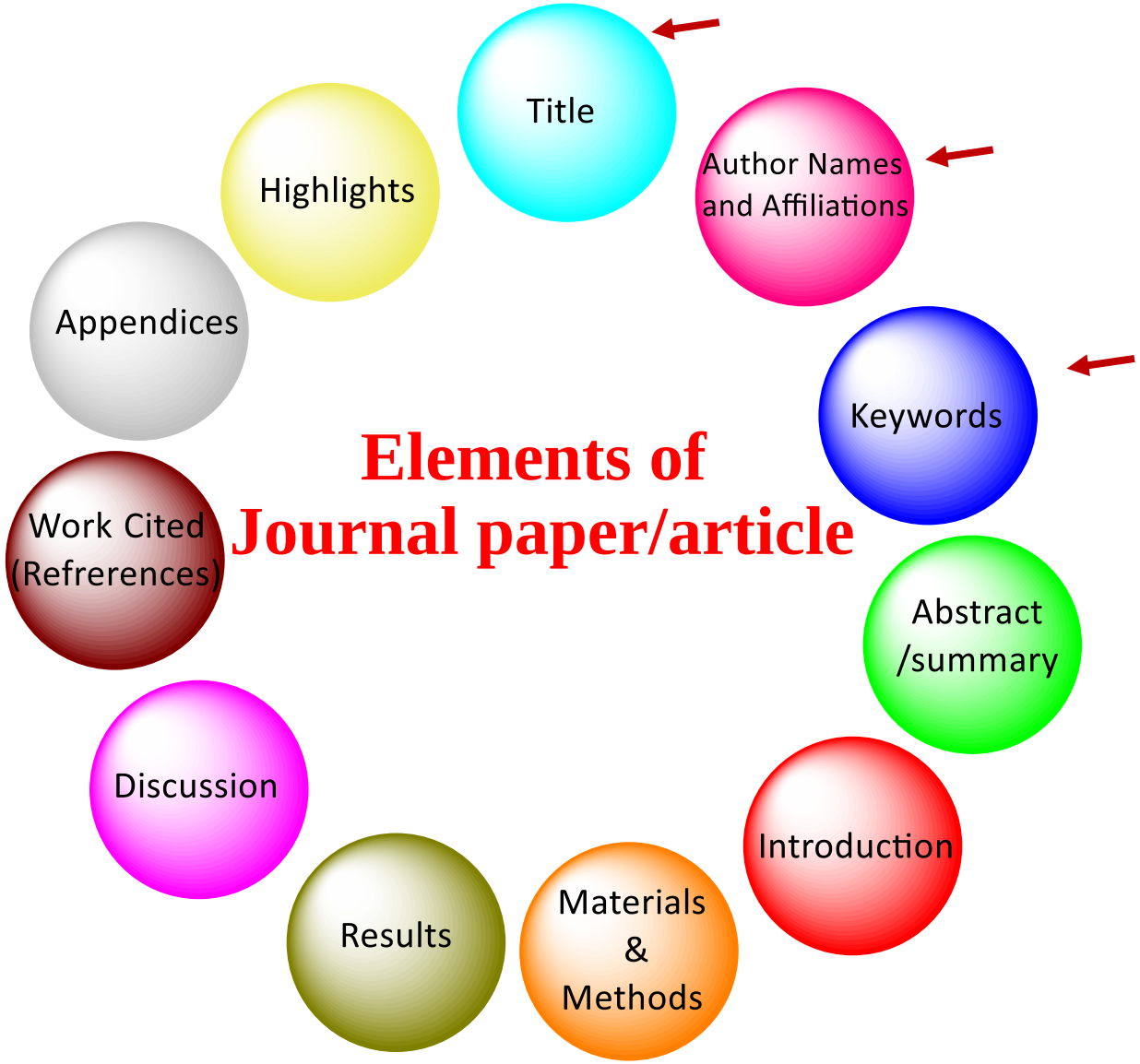
2. **Share** new research findings with the academic community

3. **Stimulate** discussion, debate, and further research on a particular topic.



# To be A Good Writer

# Elements of Journal paper/article

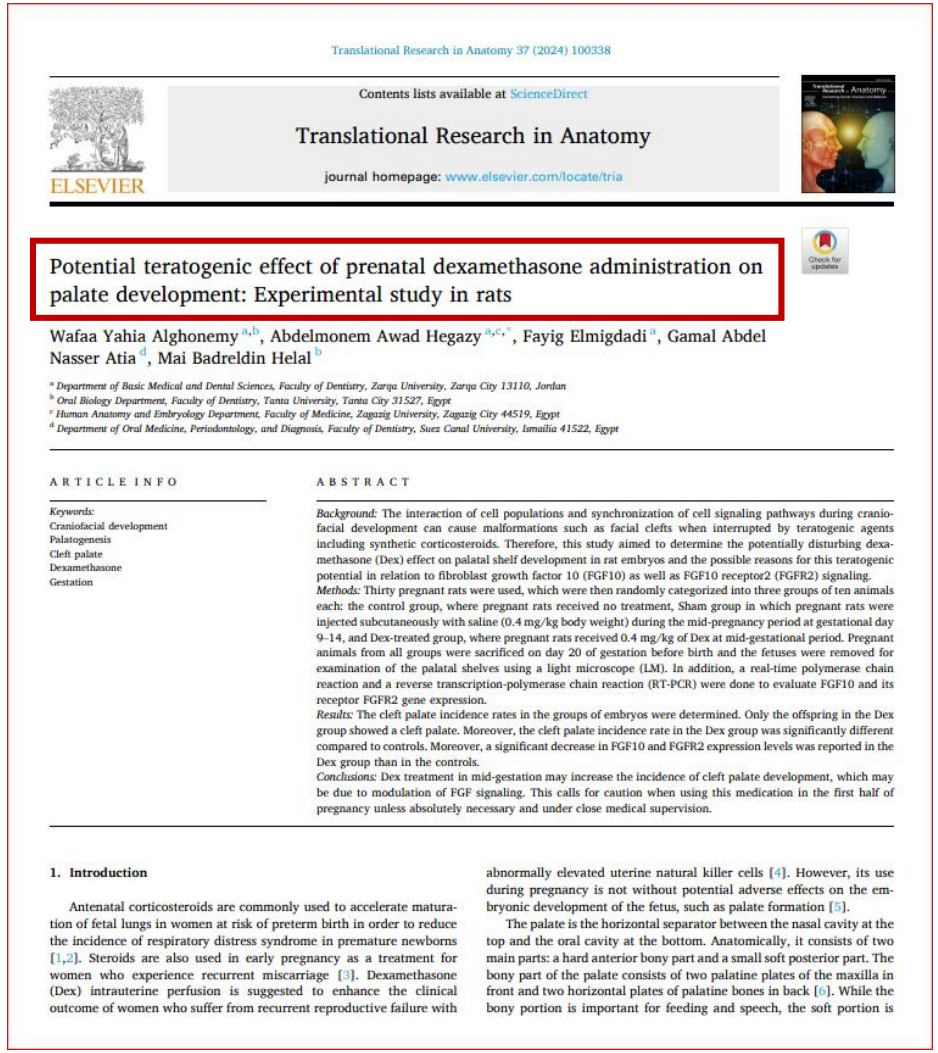


# (1) Title

**Purpose:** Provide a concise summary of the article's content

**Characteristics of a good title:**

- Clear and concise
- Understandable
- Informative: provide a clear idea of what the work is about
- Specific: avoid unclear terms
- Attention-grabbing for audience
- Free from abbreviation



Translational Research in Anatomy 37 (2024) 100338

Contents lists available at ScienceDirect

Translational Research in Anatomy

journal homepage: www.elsevier.com/locate/tria

Potential teratogenic effect of prenatal dexamethasone administration on palate development: Experimental study in rats

Wafaa Yahia Alghonemy [a,b], Abdelmonem Awad Hegazy [a,c,*], Fayig Elmigdadi [a], Gamal Abdel Nasser Atia [d], Mai Badreldin Helal [b]

[a] Department of Basic Medical and Dental Sciences, Faculty of Dentistry, Zarqa University, Zarqa City 13110, Jordan
[b] Oral Biology Department, Faculty of Dentistry, Tanta University, Tanta City 31527, Egypt
[c] Human Anatomy and Embryology Department, Faculty of Medicine, Zagazig University, Zagazig City 44519, Egypt
[d] Department of Oral Medicine, Periodontology, and Diagnosis, Faculty of Dentistry, Suez Canal University, Ismailia 41522, Egypt

ARTICLE INFO

*Keywords:*
Craniofacial development
Palatogenesis
Cleft palate
Dexamethasone
Gestation

ABSTRACT

*Background:* The interaction of cell populations and synchronization of cell signaling pathways during craniofacial development can cause malformations such as facial clefts when interrupted by teratogenic agents including synthetic corticosteroids. Therefore, this study aimed to determine the potentially disturbing dexamethasone (Dex) effect on palatal shelf development in rat embryos and the possible reasons for this teratogenic potential in relation to fibroblast growth factor 10 (FGF10) as well as FGF10 receptor2 (FGFR2) signaling.
*Methods:* Thirty pregnant rats were used, which were then randomly categorized into three groups of ten animals each: the control group, where pregnant rats received no treatment, Sham group in which pregnant rats were injected subcutaneously with saline (0.4 mg/kg body weight) during the mid-pregnancy period at gestational day 9–14, and Dex-treated group, where pregnant rats received 0.4 mg/kg of Dex at mid-gestational period. Pregnant animals from all groups were sacrificed on day 20 of gestation before birth and the fetuses were removed for examination of the palatal shelves using a light microscope (LM). In addition, a real-time polymerase chain reaction and a reverse transcription-polymerase chain reaction (RT-PCR) were done to evaluate FGF10 and its receptor FGFR2 gene expression.
*Results:* The cleft palate incidence rates in the groups of embryos were determined. Only the offspring in the Dex group showed a cleft palate. Moreover, the cleft palate incidence rate in the Dex group was significantly different compared to controls. Moreover, a significant decrease in FGF10 and FGFR2 expression levels was reported in the Dex group than in the controls.
*Conclusions:* Dex treatment in mid-gestation may increase the incidence of cleft palate development, which may be due to modulation of FGF signaling. This calls for caution when using this medication in the first half of pregnancy unless absolutely necessary and under close medical supervision.

**1. Introduction**

Antenatal corticosteroids are commonly used to accelerate maturation of fetal lungs in women at risk of preterm birth in order to reduce the incidence of respiratory distress syndrome in premature newborns [1,2]. Steroids are also used in early pregnancy as a treatment for women who experience recurrent miscarriage [3]. Dexamethasone (Dex) intrauterine perfusion is suggested to enhance the clinical outcome of women who suffer from recurrent reproductive failure with abnormally elevated uterine natural killer cells [4]. However, its use during pregnancy is not without potential adverse effects on the embryonic development of the fetus, such as palate formation [5].

The palate is the horizontal separator between the nasal cavity at the top and the oral cavity at the bottom. Anatomically, it consists of two main parts: a hard anterior bony part and a small soft posterior part. The bony part of the palate consists of two palatine plates of the maxilla in front and two horizontal plates of palatine bones in back [6]. While the bony portion is important for feeding and speech, the soft portion is

# Types of Titles

**1. Descriptive (Neutral) title:** provides a **clear summary of the content**

Example: "The Effects of Climate Change on Biodiversity“

**2. Declarative title:** makes a statement or declaration generally **describing the results**

Example: "Climate Change is the Greatest Threat to Global Biodiversity“

**3. Interrogative title:** asks a **question**

Example: "Can We Save the World's Biodiversity from Climate Change?“

**4. Comparative title:** **compares two or more** things

Example: "A Comparison of the Effects of Climate Change on Biodiversity in Different Regions"

# Tips for writing a good title

1. Use **keywords** that are relevant to the content and will help readers find the work

2. **Avoid** using technical jargon or **complex terminology** that may confuse readers

3. Keep the title **concise** and avoid using unnecessary words

4. Make the title **attractive** for the audience to encourage readers to read on

5. Use **action verbs** like "investigate,“ "analyze," or "compare" to make the title more dynamic

# Examples of good titles

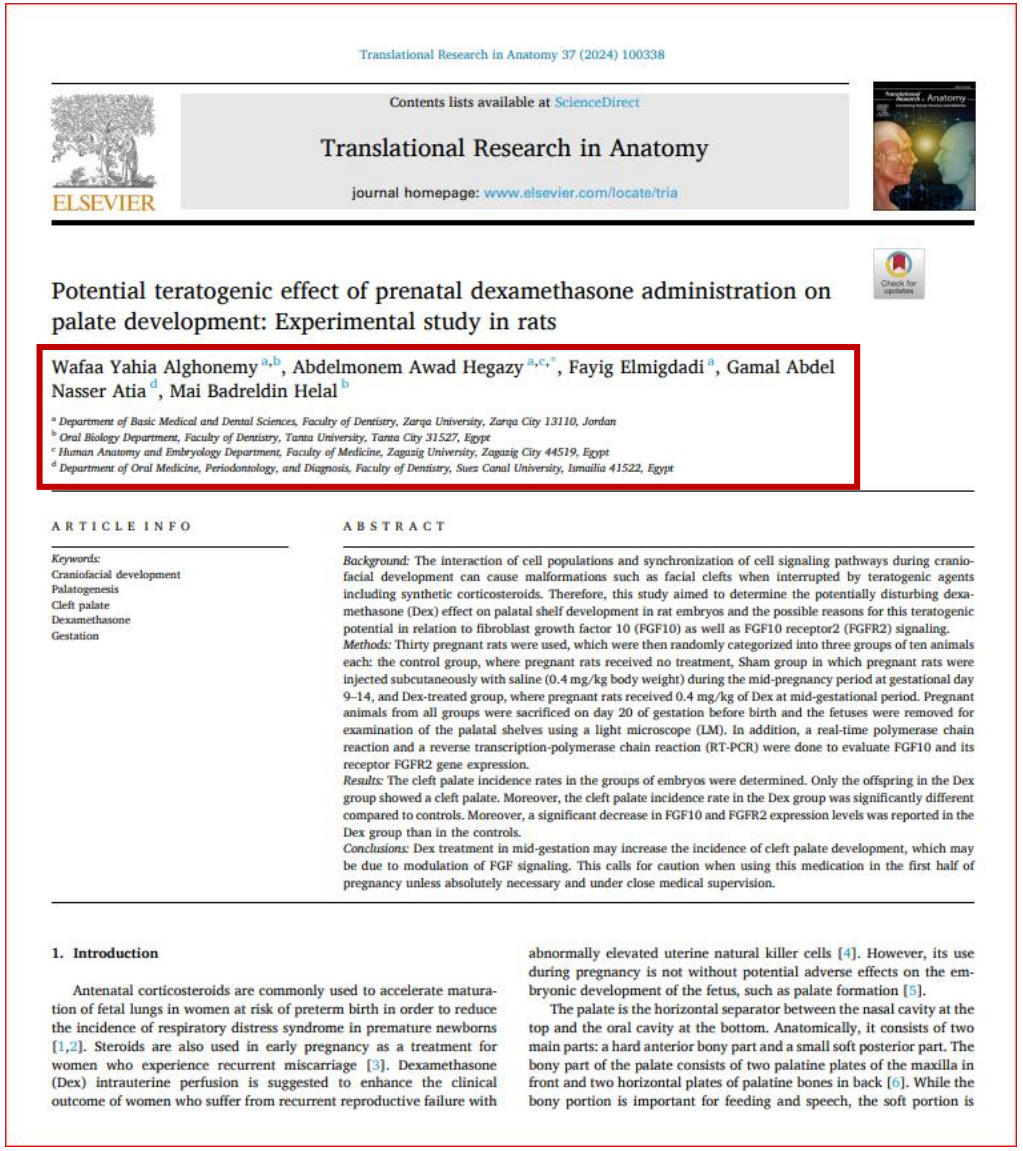
- **Example 1**: "The Impact of Social Media on Dental Health: A Systematic Review“

- **Example 2**: "Climate Change and Its Effects on Global Food Security: A Review of the Literature“

- **Example 3**: "An Investigation into the Effects of Exercise on Mental Function in Older Adults"

# (2) Authors names and Affiliation

- Each **author's name** is followed by the name of their **affiliated department**, school, university, research institute, company, or independent scholar status.
- The **primary affiliation** for each author should be the institution where the majority of their work was done.
- If an author has subsequently moved, the **current address** may additionally be stated.
- Please ensure that the **names of all authors** are present and **correctly spelled** and that addresses and affiliations are current.



Translational Research in Anatomy 37 (2024) 100338

Contents lists available at ScienceDirect

Translational Research in Anatomy

journal homepage: www.elsevier.com/locate/tria

Potential teratogenic effect of prenatal dexamethasone administration on palate development: Experimental study in rats

Wafaa Yahia Alghonemy [a,b], Abdelmonem Awad Hegazy [a,c,*], Fayig Elmigdadi [a], Gamal Abdel Nasser Atia [d], Mai Badreldin Helal [b]

[a] Department of Basic Medical and Dental Sciences, Faculty of Dentistry, Zarqa University, Zarqa City 13110, Jordan
[b] Oral Biology Department, Faculty of Dentistry, Tanta University, Tanta City 31527, Egypt
[c] Human Anatomy and Embryology Department, Faculty of Medicine, Zagazig University, Zagazig City 44519, Egypt
[d] Department of Oral Medicine, Periodontology, and Diagnosis, Faculty of Dentistry, Suez Canal University, Ismailia 41522, Egypt

ARTICLE INFO

*Keywords:*
Craniofacial development
Palatogenesis
Cleft palate
Dexamethasone
Gestation

ABSTRACT

*Background:* The interaction of cell populations and synchronization of cell signaling pathways during craniofacial development can cause malformations such as facial clefts when interrupted by teratogenic agents including synthetic corticosteroids. Therefore, this study aimed to determine the potentially disturbing dexamethasone (Dex) effect on palatal shelf development in rat embryos and the possible reasons for this teratogenic potential in relation to fibroblast growth factor 10 (FGF10) as well as FGF10 receptor2 (FGFR2) signaling.
*Methods:* Thirty pregnant rats were used, which were then randomly categorized into three groups of ten animals each: the control group, where pregnant rats received no treatment, Sham group in which pregnant rats were injected subcutaneously with saline (0.4 mg/kg body weight) during the mid-pregnancy period at gestational day 9–14, and Dex-treated group, where pregnant rats received 0.4 mg/kg of Dex at mid-gestational period. Pregnant animals from all groups were sacrificed on day 20 of gestation before birth and the fetuses were removed for examination of the palatal shelves using a light microscope (LM). In addition, a real-time polymerase chain reaction and a reverse transcription-polymerase chain reaction (RT-PCR) were done to evaluate FGF10 and its receptor FGFR2 gene expression.
*Results:* The cleft palate incidence rates in the groups of embryos were determined. Only the offspring in the Dex group showed a cleft palate. Moreover, the cleft palate incidence rate in the Dex group was significantly different compared to controls. Moreover, a significant decrease in FGF10 and FGFR2 expression levels was reported in the Dex group than in the controls.
*Conclusions:* Dex treatment in mid-gestation may increase the incidence of cleft palate development, which may be due to modulation of FGF signaling. This calls for caution when using this medication in the first half of pregnancy unless absolutely necessary and under close medical supervision.

**1. Introduction**

Antenatal corticosteroids are commonly used to accelerate maturation of fetal lungs in women at risk of preterm birth in order to reduce the incidence of respiratory distress syndrome in premature newborns [1,2]. Steroids are also used in early pregnancy as a treatment for women who experience recurrent miscarriage [3]. Dexamethasone (Dex) intrauterine perfusion is suggested to enhance the clinical outcome of women who suffer from recurrent reproductive failure with abnormally elevated uterine natural killer cells [4]. However, its use during pregnancy is not without potential adverse effects on the embryonic development of the fetus, such as palate formation [5].

The palate is the horizontal separator between the nasal cavity at the top and the oral cavity at the bottom. Anatomically, it consists of two main parts: a hard anterior bony part and a small soft posterior part. The bony part of the palate consists of two palatine plates of the maxilla in front and two horizontal plates of palatine bones in back [6]. While the bony portion is important for feeding and speech, the soft portion is

# Affiliation

- Please note that author names will be **published exactly** as they appear on the accepted submission.
- **Adding and/or deleting authors** at the revision stage is generally not permitted, but it may be necessary in some cases.
- The **reasons** for these changes in authorship should be **explained**.
- Approval of the change during revision is at the discretion of the **Editor-in-Chief.**
- Please note that journals may have **individual policies** on adding and/or deleting authors during the revision stage.

# Affiliation guidelines

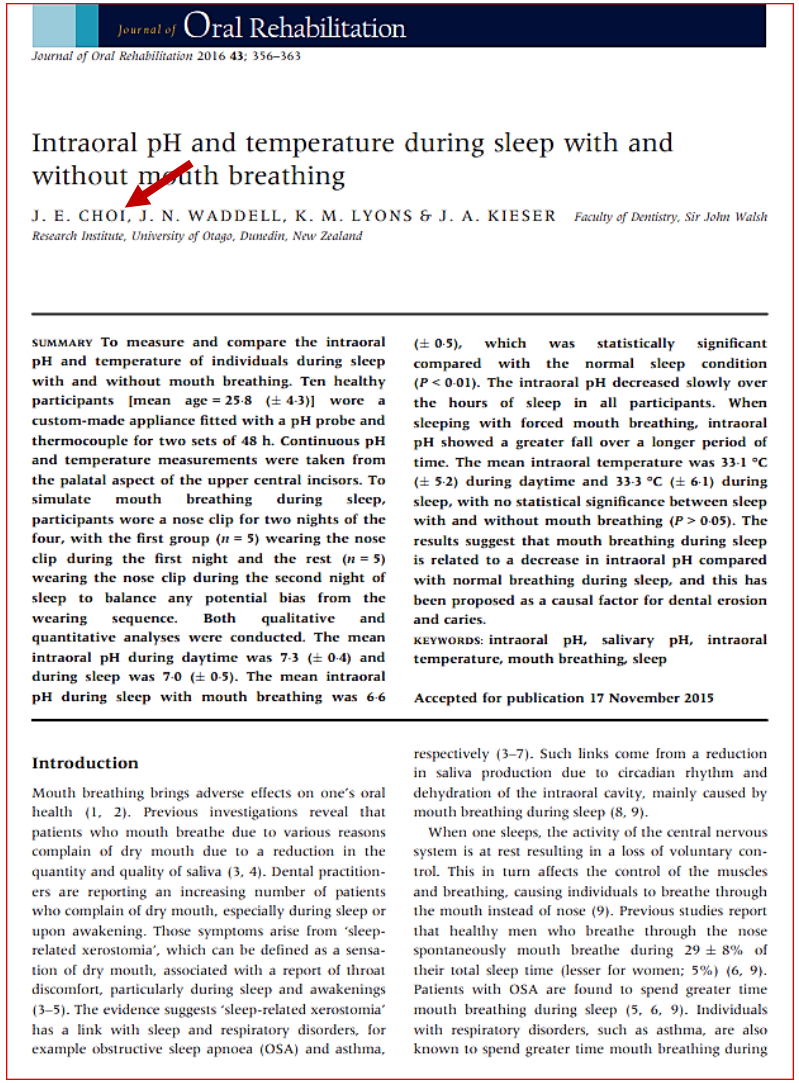
## 1) Publish under a Consistent Author Name

- It is recommended that you publish under a consistent name format and try to make that name format **as unique as possible**.
- Ensure that all future publications list your name in the **same format**, including those in which you are **a co-author**.
- For example, including a **middle name** or initial such as "H. C." can help differentiate "Martin H. C. Chen" from the approximately 30 other "Martin Chen"s in the Web of Science database

## 2) Include Author Identifiers such as ORCID. Whenever possible,

it is recommended that you submit an **author identifier** when you submit a manuscript.

## 3) Ensure Complete and Proper Listing of Affiliations



Journal of Oral Rehabilitation

*Journal of Oral Rehabilitation* 2016 **43**; 356–363

Intraoral pH and temperature during sleep with and without mouth breathing

J. E. CHOI, J. N. WADDELL, K. M. LYONS & J. A. KIESER *Faculty of Dentistry, Sir John Walsh Research Institute, University of Otago, Dunedin, New Zealand*

SUMMARY To measure and compare the intraoral pH and temperature of individuals during sleep with and without mouth breathing. Ten healthy participants [mean age = 25·8 (± 4·3)] wore a custom-made appliance fitted with a pH probe and thermocouple for two sets of 48 h. Continuous pH and temperature measurements were taken from the palatal aspect of the upper central incisors. To simulate mouth breathing during sleep, participants wore a nose clip for two nights of the four, with the first group ($n$ = 5) wearing the nose clip during the first night and the rest ($n$ = 5) wearing the nose clip during the second night of sleep to balance any potential bias from the wearing sequence. Both qualitative and quantitative analyses were conducted. The mean intraoral pH during daytime was 7·3 (± 0·4) and during sleep was 7·0 (± 0·5). The mean intraoral pH during sleep with mouth breathing was 6·6 (± 0·5), which was statistically significant compared with the normal sleep condition ($P$ < 0·01). The intraoral pH decreased slowly over the hours of sleep in all participants. When sleeping with forced mouth breathing, intraoral pH showed a greater fall over a longer period of time. The mean intraoral temperature was 33·1 °C (± 5·2) during daytime and 33·3 °C (± 6·1) during sleep, with no statistical significance between sleep with and without mouth breathing ($P$ > 0·05). The results suggest that mouth breathing during sleep is related to a decrease in intraoral pH compared with normal breathing during sleep, and this has been proposed as a causal factor for dental erosion and caries.

KEYWORDS: intraoral pH, salivary pH, intraoral temperature, mouth breathing, sleep

Accepted for publication 17 November 2015

**Introduction**

Mouth breathing brings adverse effects on one's oral health (1, 2). Previous investigations reveal that patients who mouth breathe due to various reasons complain of dry mouth due to a reduction in the quantity and quality of saliva (3, 4). Dental practitioners are reporting an increasing number of patients who complain of dry mouth, especially during sleep or upon awakening. Those symptoms arise from 'sleep-related xerostomia', which can be defined as a sensation of dry mouth, associated with a report of throat discomfort, particularly during sleep and awakenings (3–5). The evidence suggests 'sleep-related xerostomia' has a link with sleep and respiratory disorders, for example obstructive sleep apnoea (OSA) and asthma, respectively (3–7). Such links come from a reduction in saliva production due to circadian rhythm and dehydration of the intraoral cavity, mainly caused by mouth breathing during sleep (8, 9).

When one sleeps, the activity of the central nervous system is at rest resulting in a loss of voluntary control. This in turn affects the control of the muscles and breathing, causing individuals to breathe through the mouth instead of nose (9). Previous studies report that healthy men who breathe through the nose spontaneously mouth breathe during 29 ± 8% of their total sleep time (lesser for women; 5%) (6, 9). Patients with OSA are found to spend greater time mouth breathing during sleep (5, 6, 9). Individuals with respiratory disorders, such as asthma, are also known to spend greater time mouth breathing during

# ORCID ID

- **The ORCID ID**: a unique, persistent identifier free of charge to researchers.
- All people who participate in research, scholarship, and innovation are uniquely identified and connected to their contributions across disciplines, borders, and time.

Salem *et al. BMC Oral Health* (2025) 25:17
https://doi.org/10.1186/s12903-024-05285-4

BMC Oral Health

RESEARCH Open Access

## Comparative study of pulpal response following direct pulp capping using synthesized fluorapatite and hydroxyapatite nanoparticles

Eman M. Salem[1*], Omnia M. Abdelfatah[1], Rania A. Hanafy[2], Rehab M. El-Sharkawy[3], Ghadir Elnawawy[4] and Wafaa Yahia Alghonemy[5,6]

**Abstract**

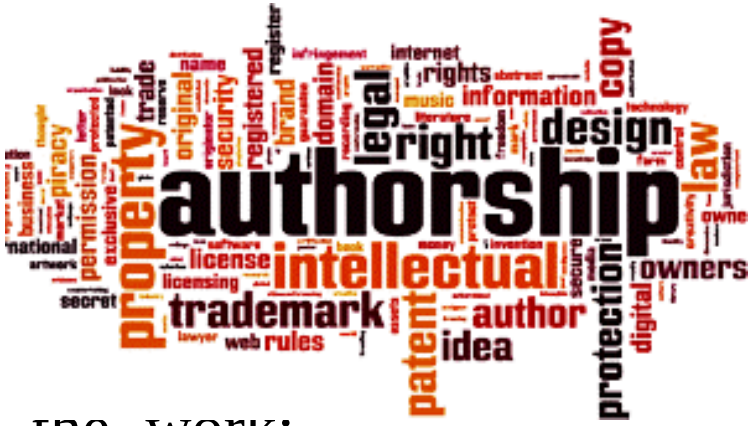
**Objective** This study aimed to investigate and compare the histological response of rabbit dental pulp after direct pulp capping with 3 different materials: mineral trioxide aggregate (MTA), nanoparticles of fluorapatite (Nano-FA), and nanoparticles of hydroxyapatite (Nano-HA) after 4 and 6-week time intervals.

**Material and methods** A total of 72 upper and lower incisor teeth from 18 rabbits were randomly categorized into 3 groups)24 incisors from six rabbits each. MTA Group: teeth were capped with MTA. Nano-FA Group: teeth were capped with fluorapatite nanoparticles. Nano-HA Group: teeth were capped with hydroxyapatite nanoparticles. Blood samples were collected to examine some antioxidant enzymes nitric oxide (NO), superoxide dismutase (SOD), catalase (CAT), glutathione peroxidase (GPx), glutathione (GSH), interleukin-6 (IL-6), and tumor necrosis factor-α (TNF-α). After that, three rabbits from each group were euthanized after 4 and 6 weeks, respectively. Pulp tissues of all teeth in all groups were histologically observed.

**Results** The obtained results showed that both Nano-HA induced the formation of thick dentin bridges with irregular dentin patterns at 6 weeks, while MTA and Nano-FA induced no dentin bridge with no tubular dentin pattern. Blood examination at the two intervals revealed no significant increase or decrease in the values of NO, SOD, CAT, GPx, GSH, and TNF-α. However, there was a significant increase in *p*-values of IL-6 in the Nano-FA treated group compared to both MTA and Nano-HA treated groups at the two intervals. Regarding the inflammatory reaction of the dental pulp, the MTA and Nano-HA groups displayed moderate inflammation, followed by Nano-FA, which showed the highest prevalence of nonpathological inflammation. Histological results were consistent with the blood examination. After 4 weeks, the Nano-FA and Nano-HA groups showed pulp fibrosis at the operating site, but the MTA showed only granulation tissues. Plus, dilated blood vessels appeared in the Nano-FA group. After 6 weeks, MTA and Nano-FA groups showed pulp fibrosis at the operating site with the persistence of dilated blood vessels with Nano-FA. The nano-HA group showed dentin bridge formation at the operating site.

**Conclusion** MTA and Nano-HA could be considered favorable materials for direct pulp capping, while Nano-FA produces nonpathological inflammatory cell reactions. Moreover, the Nano-HA was the best in dentin bridge formation. Although nano-FA increased the operating site closure, it was noticed that it significantly increased IL-6 compared to MTA at the two intervals and significantly increased IL-6 compared to Nano-HA at 6 weeks, which may

# Authorship Definition




- All authors whose names **appear on the submission**
- They made **considerable contributions** to the design of the work; analysis or interpretation of data; or the creation of new software used in the work;
- They **drafted** the work or **revised** it critically for important intellectual content;
- They **approved** the version to be published, and
- They agree to be **answerable for all aspects** of the work and ensure that questions related to the work are properly investigated and resolved.

# (3) Keywords

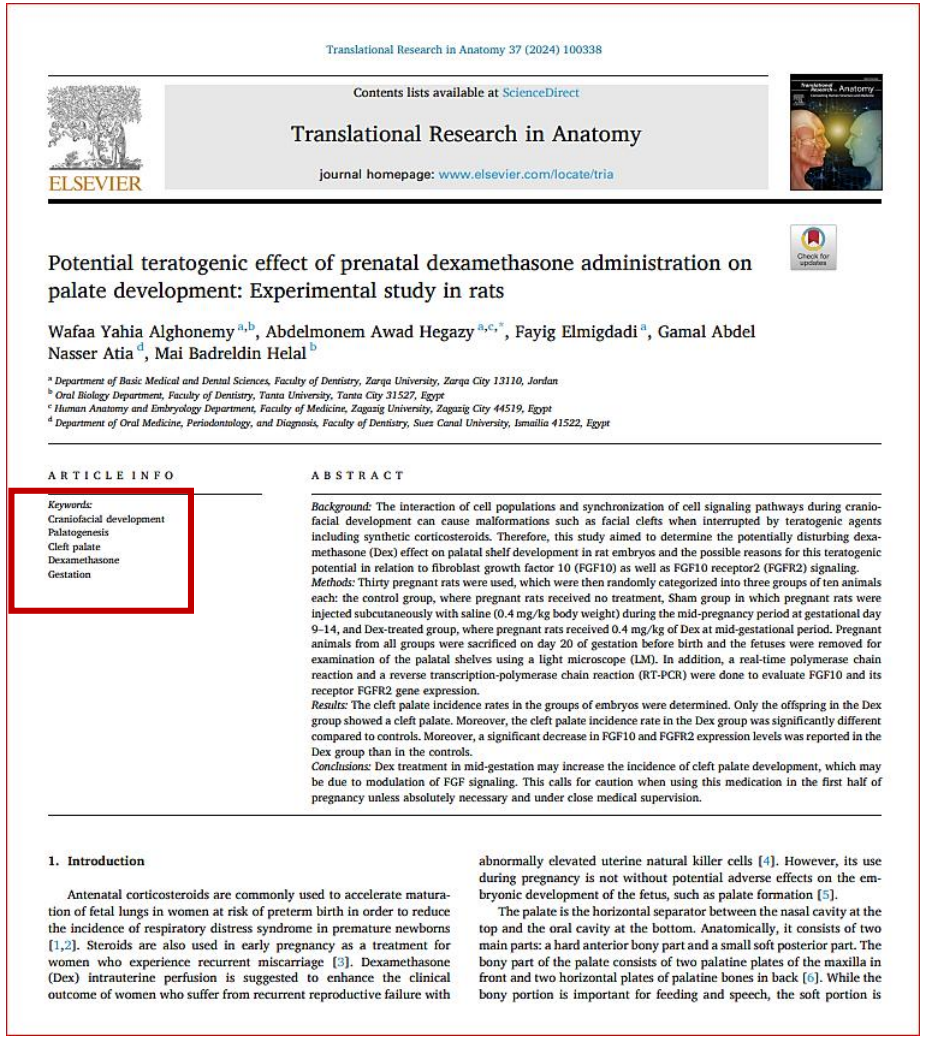
**Definition:** Keywords are words or short phrases that capture the **key aspects** of your manuscript.

## What are keywords, and why are they important?

They serve two main purposes:

1. **Categorizing** your paper during the submission/publication process

2. **Indexing** your paper after publication



Translational Research in Anatomy 37 (2024) 100338

Contents lists available at ScienceDirect

Translational Research in Anatomy

journal homepage: www.elsevier.com/locate/tria

Potential teratogenic effect of prenatal dexamethasone administration on palate development: Experimental study in rats

Wafaa Yahia Alghonemy [a,b], Abdelmonem Awad Hegazy [a,c,*], Fayig Elmigdadi [a], Gamal Abdel Nasser Atia [d], Mai Badreldin Helal [b]

[a] Department of Basic Medical and Dental Sciences, Faculty of Dentistry, Zarqa University, Zarqa City 13110, Jordan
[b] Oral Biology Department, Faculty of Dentistry, Tanta University, Tanta City 31527, Egypt
[c] Human Anatomy and Embryology Department, Faculty of Medicine, Zagazig University, Zagazig City 44519, Egypt
[d] Department of Oral Medicine, Periodontology, and Diagnosis, Faculty of Dentistry, Suez Canal University, Ismailia 41522, Egypt

ARTICLE INFO

*Keywords:*
Craniofacial development
Palatogenesis
Cleft palate
Dexamethasone
Gestation

ABSTRACT

*Background:* The interaction of cell populations and synchronization of cell signaling pathways during craniofacial development can cause malformations such as facial clefts when interrupted by teratogenic agents including synthetic corticosteroids. Therefore, this study aimed to determine the potentially disturbing dexamethasone (Dex) effect on palatal shelf development in rat embryos and the possible reasons for this teratogenic potential in relation to fibroblast growth factor 10 (FGF10) as well as FGF10 receptor2 (FGFR2) signaling.
*Methods:* Thirty pregnant rats were used, which were then randomly categorized into three groups of ten animals each: the control group, where pregnant rats received no treatment, Sham group in which pregnant rats were injected subcutaneously with saline (0.4 mg/kg body weight) during the mid-pregnancy period at gestational day 9–14, and Dex-treated group, where pregnant rats received 0.4 mg/kg of Dex at mid-gestational period. Pregnant animals from all groups were sacrificed on day 20 of gestation before birth and the fetuses were removed for examination of the palatal shelves using a light microscope (LM). In addition, a real-time polymerase chain reaction and a reverse transcription-polymerase chain reaction (RT-PCR) were done to evaluate FGF10 and its receptor FGFR2 gene expression.
*Results:* The cleft palate incidence rates in the groups of embryos were determined. Only the offspring in the Dex group showed a cleft palate. Moreover, the cleft palate incidence rate in the Dex group was significantly different compared to controls. Moreover, a significant decrease in FGF10 and FGFR2 expression levels was reported in the Dex group than in the controls.
*Conclusions:* Dex treatment in mid-gestation may increase the incidence of cleft palate development, which may be due to modulation of FGF signaling. This calls for caution when using this medication in the first half of pregnancy unless absolutely necessary and under close medical supervision.

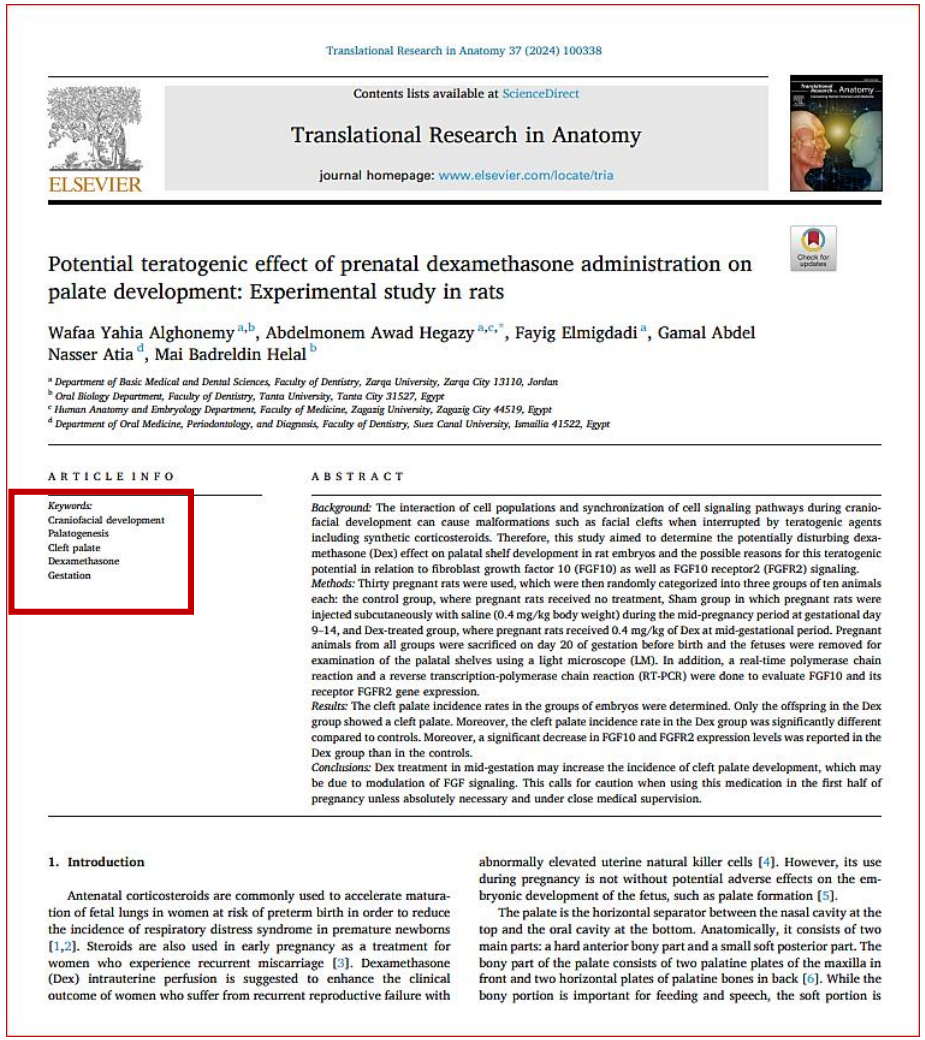
1. Introduction

Antenatal corticosteroids are commonly used to accelerate maturation of fetal lungs in women at risk of preterm birth in order to reduce the incidence of respiratory distress syndrome in premature newborns [1,2]. Steroids are also used in early pregnancy as a treatment for women who experience recurrent miscarriage [3]. Dexamethasone (Dex) intrauterine perfusion is suggested to enhance the clinical outcome of women who suffer from recurrent reproductive failure with abnormally elevated uterine natural killer cells [4]. However, its use during pregnancy is not without potential adverse effects on the embryonic development of the fetus, such as palate formation [5].

The palate is the horizontal separator between the nasal cavity at the top and the oral cavity at the bottom. Anatomically, it consists of two main parts: a hard anterior bony part and a small soft posterior part. The bony part of the palate consists of two palatine plates of the maxilla in front and two horizontal plates of palatine bones in back [6]. While the bony portion is important for feeding and speech, the soft portion is

# Keywords

- ✓ **Follow** the journal guidelines when selecting keywords;
- ✓ Focus on the **main topic** of your research;
- ✓ Include your techniques and/or specific **methodology**;
- ✓ **Avoid** keywords that are only **one word**;
- ✓ **Avoid overlapping** keywords in your title and those in your keyword list;
- ✓ Perform keyword research **before submitting** your article.



Translational Research in Anatomy 37 (2024) 100338

Contents lists available at ScienceDirect

Translational Research in Anatomy

journal homepage: www.elsevier.com/locate/tria

Potential teratogenic effect of prenatal dexamethasone administration on palate development: Experimental study in rats

Wafaa Yahia Alghonemy [a,b], Abdelmonem Awad Hegazy [a,c,*], Fayig Elmigdadi [a], Gamal Abdel Nasser Atia [d], Mai Badreldin Helal [b]

[a] Department of Basic Medical and Dental Sciences, Faculty of Dentistry, Zarqa University, Zarqa City 13110, Jordan
[b] Oral Biology Department, Faculty of Dentistry, Tanta University, Tanta City 31527, Egypt
[c] Human Anatomy and Embryology Department, Faculty of Medicine, Zagazig University, Zagazig City 44519, Egypt
[d] Department of Oral Medicine, Periodontology, and Diagnosis, Faculty of Dentistry, Suez Canal University, Ismailia 41522, Egypt

ARTICLE INFO

*Keywords:*
Craniofacial development
Palatogenesis
Cleft palate
Dexamethasone
Gestation

ABSTRACT

*Background:* The interaction of cell populations and synchronization of cell signaling pathways during craniofacial development can cause malformations such as facial clefts when interrupted by teratogenic agents including synthetic corticosteroids. Therefore, this study aimed to determine the potentially disturbing dexamethasone (Dex) effect on palatal shelf development in rat embryos and the possible reasons for this teratogenic potential in relation to fibroblast growth factor 10 (FGF10) as well as FGF10 receptor2 (FGFR2) signaling.
*Methods:* Thirty pregnant rats were used, which were then randomly categorized into three groups of ten animals each: the control group, where pregnant rats received no treatment, Sham group in which pregnant rats were injected subcutaneously with saline (0.4 mg/kg body weight) during the mid-pregnancy period at gestational day 9–14, and Dex-treated group, where pregnant rats received 0.4 mg/kg of Dex at mid-gestational period. Pregnant animals from all groups were sacrificed on day 20 of gestation before birth and the fetuses were removed for examination of the palatal shelves using a light microscope (LM). In addition, a real-time polymerase chain reaction and a reverse transcription-polymerase chain reaction (RT-PCR) were done to evaluate FGF10 and its receptor FGFR2 gene expression.
*Results:* The cleft palate incidence rates in the groups of embryos were determined. Only the offspring in the Dex group showed a cleft palate. Moreover, the cleft palate incidence rate in the Dex group was significantly different compared to controls. Moreover, a significant decrease in FGF10 and FGFR2 expression levels was reported in the Dex group than in the controls.
*Conclusions:* Dex treatment in mid-gestation may increase the incidence of cleft palate development, which may be due to modulation of FGF signaling. This calls for caution when using this medication in the first half of pregnancy unless absolutely necessary and under close medical supervision.

1. Introduction

Antenatal corticosteroids are commonly used to accelerate maturation of fetal lungs in women at risk of preterm birth in order to reduce the incidence of respiratory distress syndrome in premature newborns [1,2]. Steroids are also used in early pregnancy as a treatment for women who experience recurrent miscarriage [3]. Dexamethasone (Dex) intrauterine perfusion is suggested to enhance the clinical outcome of women who suffer from recurrent reproductive failure with abnormally elevated uterine natural killer cells [4]. However, its use during pregnancy is not without potential adverse effects on the embryonic development of the fetus, such as palate formation [5].

The palate is the horizontal separator between the nasal cavity at the top and the oral cavity at the bottom. Anatomically, it consists of two main parts: a hard anterior bony part and a small soft posterior part. The bony part of the palate consists of two palatine plates of the maxilla in front and two horizontal plates of palatine bones in back [6]. While the bony portion is important for feeding and speech, the soft portion is

# Keywords Example

**Manuscript title:** Incidence of major depressive disorder after coronary artery bypass grafting in young-old women

**Poor keywords:** cardiology, women, MDD [an abbreviation]


**Good keywords:** depression, coronary disease
THANK
YOU